



Soil and Rock Logging Explanatory Notes and Abbreviations

AS1726-2017

Soil and rock descriptions on the engineering log sheets are generally in accordance with AS1726-2017 Geotechnical Site Investigations. For full details of the logging standards refer to Section 6 of AS1726-2017. A summary of the sequence of descriptive terms used to describe soil and rock on the engineering log sheets is outlined below.

Soil Description

SOIL NAME: plasticity or particle characteristics of major component; colour; structure; secondary and other minor components. The AS1726 group symbol; consistency/density; and moisture condition are listed as abbreviations in separate columns. Geological origin and additional observations as required such as soil origin i.e., FILL, ALLUVIUM and other significant details are recorded in a separate column.

The major soil component (capitalised) is assessed after removal of cobble size or larger components (>63 mm), soils containing 35% or more passing the 75 micron sieve are classified as fine grained soils. Fine grained soils are classified on the behaviour of the fine grained component, not percentage of silt and clay.

Rock Description

ROCK NAME: grain size and type; colour; fabric and texture; structure; minor components; and bedding dip. The geological formation; rock strength; weathering/alteration; mass defect spacing; and defect descriptions are listed under separate columns.

Field Samples and In Situ Tests

Field samples and tests are recorded in the relevant column using abbreviations described under General Symbols and Abbreviations.

Field tests have been used to assess soil consistency/density and rock strength, and unless specifically stated otherwise, have been transferred directly to the engineering log sheets. Field descriptions are updated based on laboratory test results and presented on the log.

Organic and Artificial Material

Soil containing greater than 25% organic material is described as PEAT. Soils containing between 2% and 25% organic material are prefixed by "Organic" (e.g., Organic CLAY). TOPSOIL is a specific soil origin that may also contain both living and dead organic materials. Organic matter is described using terms such as fibrous peat, charcoal, wood fragments, roots (>2 mm diameter) or root fibres (<2 mm diameter)

Any material containing evidence that it has been placed by man-made mechanisms, including compacted embankment materials and materials containing evidence of artificial materials is prefixed as FILL. Waste fill is described using terms such as domestic refuse, oil, bitumen, brickbats, concrete rubble, fibrous plaster, wood pieces, wood shavings, sawdust, iron filings, drums, steel bars, steel scrap, bottles, broken glass, or leather.

Organic and artificial material cannot be adequately described using soil classification terms. They are mentioned, at the end of the description using qualitative terms such as "rare", "occasional" or "frequent", e.g., "SAND with rare gravel size brick fragments". These qualitative terms are relative, for which no definition of percentage is given.

Structure

The structure of soil (or rock) is usually applicable to cohesive soils or rock. Typical terms that are used on the engineering log sheets include: intact (no joints), fissured (closed joints), voided, vesicular, slickensided (sheared), interbedded, laminated and cemented.

Plasticity

Descriptive Term	Range of Liquid Limit (%)
Low plasticity	≤ 35
Medium plasticity	> 35 ≤ 50
High plasticity	> 50

Colour

Colour has been assessed in the "moist" condition using basic colours and the modifiers pale, dark and mottled. Borderline colours are described as a combination of the two colours (e.g., red-brown). When describing the colour of defect infill, the following abbreviations are used in the defect description column.

Colour abbreviations

Term	Abbreviation
Brown	br
Grey	gy
Black	bk
White	wh
Blue	bl
Green	gr
Yellow	yl
Orange	or
Red	rd
Pale	pl
Dark	dk
Mottled	mtld

Moisture Condition

Term	Description			
	Cohesive		Granular	
Dry	-		D	Non-cohesive and free running
Moist	w < PL	Hard and friable or powdery (dry)	M	Soil feels cool, darkened in colour, tends to cohere
	w ≈ PL	Soils can be moulded at a moisture content approximately equal to the plastic limit		
	w > PL	Soils usually weakened and free water forms on hands when handling		
Wet	w ≈ LL	Wet, near liquid limit	W	Soil feels cool, darkened in colour, tends to cohere, free water
	w > LL	Wet, wet of liquid limit		

Geological Origin - Soils

Term	Description	
Weathered in Place Material	Extremely weathered material	Structure and fabric of parent rock visible (described with soil terminology)
	Residual	Structure and fabric of parent rock not visible
Transported	Aeolian	Deposited by wind
	Alluvium	Deposited by streams and rivers
	Colluvium	Deposited on slopes (transported downslope)
	Lacustrine	Deposited by lakes
	Marine	Deposited in oceans, bays, beaches and estuaries
Topsoil	Mantle of surface and/or near surface soil often but not always defined by high levels of organic material (prefix to soil description)	
Fill	Soil, rock or refuse placed by humans in either controlled or uncontrolled conditions (prefix to soil description)	

NOTE: Where there is doubt to origin 'possibly' or 'probably' are used

Soil Classification

Field Identification Procedures (Excluding particles larger than 63mm and basing fractions on estimated mass)					Log Graphic	Group Symbol	Typical Names	Secondary and Minor Components	Laboratory Classification Criteria						
Coarse Grained Soils More than 65% of material less than 63 mm is larger than 0.075 mm	GRAVEL More than 50% of coarse fraction is larger than 2.36mm	clean GRAVEL ($<5\%$ fines)	Wide range in grain size and substantial amounts of all intermediate sizes, not enough fines to bind coarse grains, no dry strength			GW	Well graded GRAVEL or sandy GRAVEL	Fine Grains >12% >30% >15% <30% >5% <12% <5% Coarse Grains >30% >15% <30% >5% <12% <15% Secondary/Minor Descriptors: Prefix "sandy" or "silty" etc. Add "with sand/silt" etc. Add "trace sand/silt" etc.	Laboratory grain size curve used in identifying the fractions as given under field identification	$c_u = \frac{D_{60}}{D_{10}}$ $C_u \geq 4$ $c_c = \frac{(D_{60})^2}{D_{10} \times D_{40}}$ $C_c = 1 - 3$					
			Predominantly one size or range of sizes with some intermediate sizes missing, not enough fines to bind coarse grains, no dry strength			GP	Poorly graded GRAVEL or sandy GRAVEL			<5% fines, not meeting all gradation requirements for GW.(i.e., $C_u < 4$ or $C_c \neq 1 - 3$)					
		GRAVEL ($\geq 12\%$ fines)	'Dirty' materials with excess of non-plastic fines, zero to medium dry strength			GM	silty GRAVEL or silty sandy GRAVEL			>12% Fines with Atterberg limits below 'A' line					
			'Dirty' materials with excess of plastic fines, medium to high dry strength			GC	clayey GRAVEL or clayey sandy GRAVEL			>12% Fines with Atterberg limits above 'A' line					
	SAND More than 50% of coarse fraction is smaller than 2.36mm	Clean SAND ($<5\%$ fines)	Wide range in grain size and substantial amounts of all intermediate sizes, not enough fines to bind coarse grains, no dry strength			SW	Well graded SAND or gravelly SAND			$c_u = \frac{D_{60}}{D_{10}}$ $C_u \geq 6$ $c_c = \frac{(D_{60})^2}{D_{10} \times D_{40}}$ $C_c = 1-3$					
			Predominantly one size or range of sizes with some intermediate sizes missing, not enough fines to bind coarse grains, no dry strength			SP	Poorly graded SAND or gravelly SAND			<5% fines, not meeting all gradation requirements for SW. (i.e., $C_u < 6$ or $C_c \neq 1 - 3$)					
		SAND ($\geq 12\%$ fines)	'Dirty' materials with excess of non-plastic fines, zero to medium dry strength			SM	silty SAND			>12% Fines with Atterberg limits below 'A' line.					
			'Dirty' materials with excess of plastic fines, medium to high dry strength			SC	clayey SAND			>12% Fines with Atterberg limits above 'A' line.					
Fine Grained Soils 35% or more of material less than 63 mm is smaller than 0.075 mm	IDENTIFICATION PROCEDURES ON FRACTIONS < 0.2 mm						Coarse grained Secondary/Minor descriptors: prefix "gravelly" or "sandy" as appropriate add "with sand/gravel" as appropriate add "trace sand/gravel" as appropriate Coarse grained Secondary/Minor descriptors: prefix "gravelly" or "sandy" as appropriate add "with sand/gravel" as appropriate add "trace sand/gravel" as appropriate Coarse grained Secondary/Minor descriptors: prefix "gravelly" or "sandy" as appropriate add "with sand/gravel" as appropriate add "trace sand/gravel" as appropriate	Laboratory classification from Modified Casagrande Chart	Modified Casagrande Chart for classification of fine-grained soils NOTE: The U' line is an approximate upper bound for most natural soils. If data plots above the U line, it may represent unusual/problem soil behaviour, or unreliable data and should be considered carefully.						
	SILT	LIQUID LIMIT	DRY STRENGTH	DILATANCY	TOUGHNESS							MH	Inorganic SILT, gravelly SILT, sandy SILT		
		>50%	Low to medium	None to slow	Low to medium							ML	Inorganic SILT and very fine sand, rock flour, silty or clayey fine sand or silt.		
		$\leq 50\%$	None to low	Slow to rapid	Low							OL	Organic SILT (2%<organics<25%)		
			Low to medium	Slow	Low										
	Borderline SILT/ CLAY	$\leq 35\%$	Low to Medium	None to slow	Low to Medium							ML-CL	clayey SILT/silty CLAY		
	CLAY	>50%	High to very high	None	High							CH	Inorganic CLAY, sandy CLAY, gravelly CLAY		
		>35 and $\leq 50\%$	Medium to high	None to slow	Medium							CI			
		$\leq 35\%$									CL				
		>35%	Medium to high	None to very slow	Low to medium							OH	Organic CLAY (2%<organics<25%)		
	PEAT (>25% organic by dry weight)	Readily identified by colour, odour, spongy feel and frequently by fibrous texture										Pt	PEAT, sandy PEAT		
	Boundary classifications – soils possessing characteristics of two groups are designated by combinations of group symbols. For example: GP-GC for gravel with >5% and $\leq 12\%$ clay fines; CI-CH for clay soils where field estimates of plasticity are borderline between medium and high														
	Other log graphics: FILL, TOPSOIL, ASPHALT, CONCRETE, sandy GRAVEL, gravelly SAND, gravelly SILT, sandy SILT, gravelly CLAY, sandy CLAY														



Sedimentary and Metamorphic Rock Types, Grain Size, Defect Spacing and Planar Structure

Grain Size/ Spacing Thickness	Soil Grain Size Term	Rock Type							Defect Spacing Term	Bedding Thickness Term
		Sedimentary				Metamorphic		Igneous/ Metamorphic grain size		
		Deposited rock type	>90% carbonate (prefix IMPURE 50%-90%)		Volcanic ejecta	Foliated	Non-Foliated			
			Low porosity	Porous						
> 2m	large BOULDERS	 CONGLOMERATE (larger rounded grains in a finer matrix) or  BRECCIA (angular or irregular rock fragments in a finer matrix)	 LIMESTONE (predominantly CaCO³) or  DOLOMITE (predominantly CaMgCO³)	 CALCIRUDITE or  VOLCANIC BRECCIA (angular or irregular fragments in a finer matrix)	 GNEISS	 MARBLE (crystalline CaCO³)/  QUARTZITE (fused quartz)/	COARSE	VERY WIDE (VW)	VERY THICKLY BEDDED	
0.6 - 2m	medium BOULDERS							WIDE (W)	THICKLY BEDDED	
0.2 - 0.6 m	small BOULDERS							MEDIUM (M)	MEDIUM BEDDED	
60 - 200 mm	COBBLES							CLOSE (C)	THINLY BEDDED	
20 - 60 mm	coarse GRAVEL							VERY CLOSE (VC)	VERY THINLY BEDDED	
6 - 20 mm	medium GRAVEL							EXTREME LY CLOSE (EC)	THICKLY LAMINATED	
2 - 6 mm	fine GRAVEL							 SANDSTONE/  GREYWACKE (rock fragments)/  ARKOSE (mainly feldspar)/  QUARTZOZE SANDSTONE (quartz and siliceous cement)	 CALCARENITE	 TUFF
0.6 - 2.0 mm	coarse SAND									
0.2 - 0.6 mm	medium SAND									
0.06 - 0.2 mm	Fine SAND									
2 - 60 µm	SILT	 MUDSTONE/  SHALE (fissile)/  LAMINITE (bedded)	 SILTSTONE	 CALCISILTITE	 Fine grained TUFF	 PHYLLITE (undulose)/  SLATE (planar)	FINE			
< 2 µm	CLAY	 CLAYSTONE	 CALCILUTITE							

NOTES: EVAPORITE is sedimentary rock which consist mainly of salts e.g.: Halite (NaCl), Anhydrite and Gypsum (CaSO₄); COAL is mostly organic sedimentary rock that consists of indurated accumulations of plant debris

Classification of Igneous Rocks (Massive Crystalline)

Grain Size (mm)	Much quartz, pale (felsic)	←————→	Little quartz, dark (mafic)
COARSE (>2)	GRANITE	DIORITE	GABBRO
MEDIUM (0.06-2)	MICROGRANITE	MICRODIORITE	DOLERITE
FINE (<0.06)	RHYOLITE	ANDESITE	BASALT

NOTES: PEGMATITE consist of large crystals often forming a dyke or vein
 VOLCANIC GLASS and OBSIDIAN are rocks that have cooled quickly and have a glassy texture
 PORPHYRY is an igneous rock consisting of large crystals in a much finer matrix

Duricrust (soils cemented to rock)

FERRICRETE (iron oxide cemented)
SILCRETE (silica cemented)
GYPCRETE (salt cemented)
CALCRETE (CaCO ₃ cemented – dominated by replacement features)

No Core

NO CORE (description, defect and strength logs show beak in logging)



Grain Size

Soil Type	Fine Grained Soil		Coarse Grained Soil						Override
	Clay	Silt	Sand			Gravel			Cobbles
			Fine	Medium	Coarse	Fine	Medium	Coarse	
Grain Size	< 2 µm	2-75 µm	0.075-0.21 mm	0.21-0.6 mm	0.6-2.36 mm	2.36-6.7 mm	6.7-19 mm	19-63 mm	63-200 mm
Field Guide	Shiny, Not visible under 10x	Dull, Visible under 10x	Visible by eye	Visible at < 1 m	Visible at < 3 m	Visible at < 5 m	Road gravel	Rail ballast	Beaching

Boulders (>200 mm) and Cobbles (63-200 mm) are considered override materials and are separated from the sample prior to describing the soil. Where cobbles or boulders are present the soil description is preceded by 'MIXTURE OF SOIL AND COBBLES/BOULDERS' with the word order indicating the dominant proportion first followed by the soil description and the estimate proportion of override materials indicating whether they are supported by the soil matrix.

Grain Shape

	Angular	Sub-angular	Sub-rounded	Rounded
Equi-dimensional particles				

Essentially two-dimensional particles are described as "flaky" or "platy" while essentially one-dimensional particles are described as "elongated". Particle composition should be described where significant (e.g., "quartz sand")

Relative Density (non-cohesive soils)

Term	Very Loose	Loose	Medium Dense	Dense	Very Dense
Abbreviation	VL	L	MD	D	VD
Density Index (%)	≤ 15	> 15 ≤ 35	> 35 ≤ 65	> 65 ≤ 85	> 85
Uncorrected SPT (N) Blow count [#]	0 - 4	4 - 10	10 - 30	30 - 50	> 50
Field Guide	Ravels	Shovels easily	Shovelling very difficult	Pick required	Pick difficult

NOTE: Density Index as a measure of relative density only applies to dry cohesionless soils. Correlations between relative density and SPT vary considerably depending on grain size and angularity, overburden pressure, moisture content, fines content, cementation and SPT efficiency. The above SPT correlation is a rough field guide for clean, dry, fine to medium sands at a depth of 5 to 15 m after Gibbs & Holtz (1957). Where relative density is important to design, different correlations considering the factors discussed above should be used.

Consistency (cohesive soils)

Based on undrained strength (S_u) estimated in field from pocket penetrometer or shear vane

Term	Very Soft	Soft	Firm	Stiff	Very Stiff	Hard
Abbreviation	VS	S	F	St	VSt	H
Undrained Shear Strength (kPa)	≤ 12	> 12 ≤ 25	> 25 ≤ 50	> 50 ≤ 100	> 100 ≤ 200	> 200
Approx. SPT (N) Blow count [#]	0 - 2	2 - 4	4 - 8	8 - 15	15 - 30	> 30
Field Guide	Exudes between the fingers when squeezed	Can be moulded by light finger pressure	Can be moulded by strong finger pressure	Cannot be moulded by fingers. Can be indented by thumb	Can be indented by thumb nail	Can be indented with difficulty with thumb nail

NOTE: SPT is not a direct measure of consistency and ranges are only a rough field guide. Plasticity, fissuring, moisture, and sand and gravel content can all affect SPT values dramatically.

Rock Material Strength

Term	Soil	Very Low	Low	Medium	High	Very High	Extremely High
Abbreviation	-	VL	L	M	H	VH	EH
UCS (MPa)	≤ 0.6	> 0.6 ≤ 2	> 2 ≤ 6	> 6 ≤ 20	> 20 ≤ 60	> 60 ≤ 200	> 200
I_s (50) (MPa)	-	> 0.03 ≤ 0.1	> 0.1 ≤ 0.3	> 0.3 ≤ 1	> 1 ≤ 3	> 3 ≤ 10	> 10
Field Guide	Soil strength – logged as soil using consistency	Material crumbles under firm blow with sharp end of pick. Can be peeled with a knife. Too hard to cut a triaxial sample by hand. Pieces up to 3 cm thick can be broken by finger pressure.	Easily scored with a knife. Indentations of 1mm - 3mm in the specimen with firm blows of the pick point. Has dull sound under hammer. A piece of core 150 mm long 50 mm diameter may be broken by hand	Readily scored with a knife. A piece of core 150 mm long 50 mm diameter can be broken by hand with difficulty	A piece of core 150 mm long 50 mm diameter cannot be broken by hand but can be broken with by a pick with a single firm blow. Rock rings under hammer	Hand specimen breaks with a pick after more than one blow. Rock rings under hammer	Specimen requires many blows with a geological pick to break through intact material. Rock rings under hammer

NOTE: Refer to Section 6.2.4.1 Rock Material Strength of AS1726-2017 for additional details.



Degree of Weathering / Alteration

Weathering and alteration relate to the rock fabric. Both involve physical and chemical changes to the rock due to changes in pressure, temperature, moisture and chemical environment. Weathering is in response to changes from exposure at the earth's surface while alteration is caused by hot liquids or gasses at depth. The distinction between weathering and alteration is important because they are likely to have different distribution patterns.

Degree of Weathering / Alteration		Weathering Symbol/ Alteration Symbol		Weathering/ Alteration Description
Residual Soil		RS		Material is weathered to such an extent that it has soil properties. Mass structure and material texture and fabric of original rock are no longer visible but the soil has not been significantly transported.
Extremely Weathered / Extremely Altered		XW / XA		Material is weathered/ altered to such an extent that it has soil properties. Mass structure and material texture of original rock are still visible. Logged as "Extremely weathered/ altered material" and described as a soil.
Highly Weathered / Highly Altered	Distinctly Weathered/ Distinctly Altered	HW / HA	DW/ DA	The whole of the rock material is discoloured usually by staining or bleaching to the extent that the colour of the original rock is not recognisable. Rock strength is changed by weathering/alteration. Some primary minerals weathered to clay minerals. Porosity may be increased by leaching, or may be decreased due to deposition of weathering product/precipitation of secondary minerals in pores.
Moderately Weathered / Moderately Altered		MW / MA		The whole of the rock material is discoloured usually by staining or bleaching to the extent that the colour of the original rock is not recognisable but shows little or no change in strength from fresh rock.
Slightly Weathered / Slightly Altered		SW / SA		Rock is partially discoloured (with staining or bleaching along joints in the case of weathering) but shows little or no change in strength from fresh rock.
Fresh		FR		Rock shows no sign of decomposition of individual minerals or colour changes.

NOTES: 1. The term "extremely weathered/altered rock" as a description is misleading. The material should be described as a soil followed by "extremely weathered/altered material"
 2. DW/ DA = distinctly weathered/altered is used where it is not practical to distinguish between HW/HA and MW/MA on basis of strength change

In some instances, it may be appropriate to describe weathering as part of a rock mass classification system. Where this is used the rock mass classification including the specifics of the weathering related to each rock class is described in the body of the report.

Common Defects in Rock Masses

Defects are described in the description column in the following order, defined by abbreviations:

Type; dip/direction; planarity; roughness; infill/coating; other descriptors (e.g., colour).

e.g., P, 30/145°, PL, RF, CT CH, 1 mm, gy - indicates a parting with 30° dip, 145° dip direction, planar, rough surfaces, coating of high plasticity clay, 1 mm thick, grey

Defects with a perpendicular thickness <10 mm thick are described as partings or joints. Defects with a perpendicular thickness ≥10 mm and <100 mm are described as seams or zones. Defects ≥100 mm thick or intersecting the full width of the core for more than 100 mm are described as new material strata on the log. Healed defects are followed by "healed" in the description.

Defects can be either described individually or grouped into generalised defect sets with similar properties. For generalised defects, the number of defects and spacing within the set are noted at the end of the description (e.g., "... , x2, 30 mm spacing"). The extent to which defects are described individually or generalised depends on the nature of the core being logged.

Defect Type

Log Abbreviation	Appearance	Term	Definition
P		Parting	A surface or crack, parallel or sub-parallel to layering or planar anisotropy (e.g., bedding, cleavage, foliation or schistosity), across which the rock has little or no tensile strength. May be open or closed.
J		Joint	A surface or crack, with no apparent shear displacement across which the rock usually has little tensile strength, but which is not parallel or sub-parallel to bedding/cleavage.
S		Sheared Surface#	A near planar, curved or undulating surface which is usually smooth, polished or slickensided and which shows evidence of shear displacement.
SZ		Sheared Zone#	Zone of rock material with roughly parallel near planar, curved or undulating boundaries cut by closely spaced joints, sheared surfaces or other defects. Some defects are usually curved and intersect to divide the mass into lenticular or wedge-shaped blocks.
MB / DL / DB		Mechanical Break / Drill Lift / Handling Break	A break in rock mass not caused by natural effects.
SS		Seams	Sheared Seam# Soil seam with roughly parallel boundaries cut by closely spaced joints, sheared surfaces or other defects.
CS			Crushed Seam# Soil seam with roughly parallel boundaries composed of disorientated and usually angular fragment of the host rock substance. Some defects are usually curved and intersect to divide the mass into lenticular or wedge-shaped blocks.
IS			Infilled Seam Soil seam with distinct roughly parallel boundaries. The infill is caused by migration of soil into open joints.
EW			Extremely Weathered Seam Seam of soil substance, weathered from host rock. Often has gradational boundaries.

NOTES: # Sheared surfaces, sheared zones, sheared seams and crushed seams are generally faults in geological terms.
 A mineral growth thicker than a cemented joint (>1 mm) or an intrusive feature is described as a defect with "vein" used as a suffix.
 Healed defects are suffixed by "healed".

Defect Planarity

Abbreviation	Description
PR	planar
CU	curved
UN	undulating

Abbreviation	Description
ST	stepped
IR	irregular

NOTE: where large scale waviness is observed it should be further described

**Defect Roughness**

Abbreviation	Description
VR	Very Rough
RF	Rough
SM	Smooth
PO	Polished
SL	Slicksided

Infill/Coating

Abbreviation	Description
CN	Clean
SN	Stained
VN	Veneer (too thin to measure, may be patchy)*
CT	Coating (≤ 1 mm thick)*
OP	Open/Voided
Ca	Calcium Carbonate
Fe	Iron Oxide
Ch	Chlorite
Qz	Quartz

*NOTE: Infill/veneers/coatings of soil should use group symbols from the soil classification table, e.g. well graded sand: SW or Clay of high plasticity: CH.

Vesicularity

Abbreviation	Description	Porosity
D	Dense	Negligible
NV	Non-vesicular	<10%
SV	Slightly vesicular	10 - 20%
HV	Highly vesicular	>20%

Cementation and Duricrust

'Weakly cemented' soils are easily disaggregated by hand in air or water.

'Moderately cemented' soils require effort to disaggregate the soil by hand in air or water.

Consistent cementation throughout the material is classified as a 'Duricrust' and logged as a rock.

Duricrust Mass Grade

Grade	Term	Description
DI	Massive or Hardpan	>90% duricrust rock forms continuous framework
DII	Vuggy or Patchy	50-90% duricrust rock forming continuous framework around soil (vuggy) or rock (patchy) materials
DIII	Nodular or Fragmental	<50% cemented gravel or cobble size nodules or fragments within soil (described as soil)

Carbonate Soils and Rocks

Rocks and soils with less than approximately 50% carbonate content as indicated by weak or sporadic effervescence in the presence of 10% HCl solution are prefixed 'Calcareous'. Soils with >50% carbonate are prefixed 'Carbonate'.

Sedimentary Rocks comprising 90% or more carbonate are described using the appropriate rock type name (see rock description). Where carbonate content is 50% to <90%, the rock type is prefixed by 'IMPURE'.

Rock Core Indices

Total core recovery (TCR) and rock quality designation (RQD) are calculated over the length of a core run as defined in AS1726-2017. Solid core recovery (SCR) is calculated similarly to RQD but includes full width pieces less than 100mm long.

Field Sampling and Testing Abbreviations

Abbreviation	Description
V	Uncorrected Borehole Vane Shear (kPa) – Peak/Residual
FV/HV	Uncorrected Field Hand Vane Shear (kPa) – Peak/Residual
PP	Pocket Penetrometer (kPa)
SPT	Standard Penetration Test
N	Uncorrected SPT blow count for 300 mm
HB	SPT Hammer Bouncing

Abbreviation	Description
FPM	Field Permeability
Lu	Lugeon/Packer Test (L/m/min)
Is ₍₅₀₎ (A)	Axial Point Load Strength Index (MPa)
Is ₍₅₀₎ (D)	Diametral Point Load Strength Index (MPa)
Is ₍₅₀₎ (I)	Irregular Point Load Strength Index (MPa)
U(X)	Undisturbed Sample (X) mm diameter
UP	Undisturbed Piston Sample
C / CONCC	Core Sample / Concrete Core Sample
D	Disturbed Sample
B	Bulk Sample
ES	Environmental Sample
RQD	Rock Quality Designation (%)
SCR	Solid Core Recovery (%)
TCR	Total Core Recovery (%)
DCP	Dynamic Cone Penetration Resistance (blows/100 mm)
PSP	Perth Sand Penetrometer Resistance (blows/150 mm)
PID	Photoionization Detector

Water

Symbol	Description
	Water level (static)
	Water level (during drilling)
	Water inflow
	Water outflow
	Complete water loss

Drilling Method

Abbreviation	Description
AD	Auger Drilling*
V / T	V Bit / Tungsten Carbide Bit*
AS	Auger Screwing
WB	Wash Boring
RR	Rock Roller/Tricone
DHH	Down Hole Hammer
PD	Percussion Drilling
CT	Cable Tool
HA	Hand Auger
DT	Diatube (114mm)
NMLC	NMLC Size Core – Triple Tube (50 mm diameter)
NQ3, HQ3, PQ3	Wireline Size Core – Triple Tube (45 mm, 61 mm, 83 mm diameter)
NQ, HQ, PQ	Wireline Size Core – Double Tube (48 mm, 64 mm, 85 mm diameter)
CA	Casing Advancer
VC	Vibro Coring
SC	Sonic Coring
E	Excavator
VE	Hydro-vacuum Excavation
GP	Geoprobe Continuous Sampling

*NOTE: Drill bit symbol used as suffix to drilling method symbol, e.g., AD/V indicates auger drilling with V-bit

Drilling Support

Abbreviation	Description
U	Unsupported
C	Casing
M	Mud
W	Water